



Information Technology Institute (ITI)
Analog IC Design Track
Summer Camp 2026
Lab 08
Negative Feedback
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Table of Contents

List of figures	4
PART 1: Feedback with Behavioral OTA	5
1) Closed loop gain vs frequency.	6
• Plot V_{out} in dB for the two values of C_{IN} (4pF and 12pF)	6
• the DC gain, the bandwidth, and the unity gain frequency	6
• Compare the DC gain, BW, and GBW with hand analysis in a table.	7
• Comment	7
2) Loop gain vs frequency.	8
• Plot loop gain in dB for the two values of C_{IN}	8
• the DC loop gain, the dominant pole, and the unity gain frequency	8
• Compare DC LG and GBW with hand analysis in a table.	9
3) Gain Desensitization. Study the variation of closed loop gain with the variation of open loop gain	11
• Plot closed loop DC gain (magnitude at 10Hz, not dB) vs A_v ..	11
• Calculate the percent change in closed loop gain	11
PART 2: Feedback with Real 5T OTA	12
1 Closed loop gain vs frequency.	12
• Compare between the results you obtained here and the results in Part 1 in a table.....	13
2. Loop gain vs frequency.	14



• Compare between the results you obtained here and the results in Part 1 in a table.....	15
.3 Gain Desensitization.	16
4. Transient analysis.	17
• Case 1: Input Frequency $F_{IN} = 1 \text{ kHz}$	17
• Case 2: Input Frequency $F_{IN} = \text{Closed-Loop Bandwidth}$	19



List of figures

Figure 1:schematic_behav	5
Figure 2: simple non-inverting amplifier	5
Figure 3:Plot Vout in dB for the two values of CIN (4pF and 12pF)	6
Figure 4: A_o , BW , W_u	6
Figure 5: loop gain	8
Figure 6: closed loop gain vs. A_v	11
Figure 7: A_{cl} , BW, GBW vs. freq	12
Figure 8:closed loop gain vs. frequency	12
Figure 9:loop gain vs. freq.	14
Figure 10:LG, BW, GBW.....	14
Figure 11:loop gain and closed loop vs. temp.....	16
Figure 12: V_{in} and V_{out} vs. time.....	17
Figure 13: V_{diff} vs. time	18
Figure 14: V_{in} , V_{out} and V_{diff} vs. time	19



PART 1: Feedback with Behavioral OTA

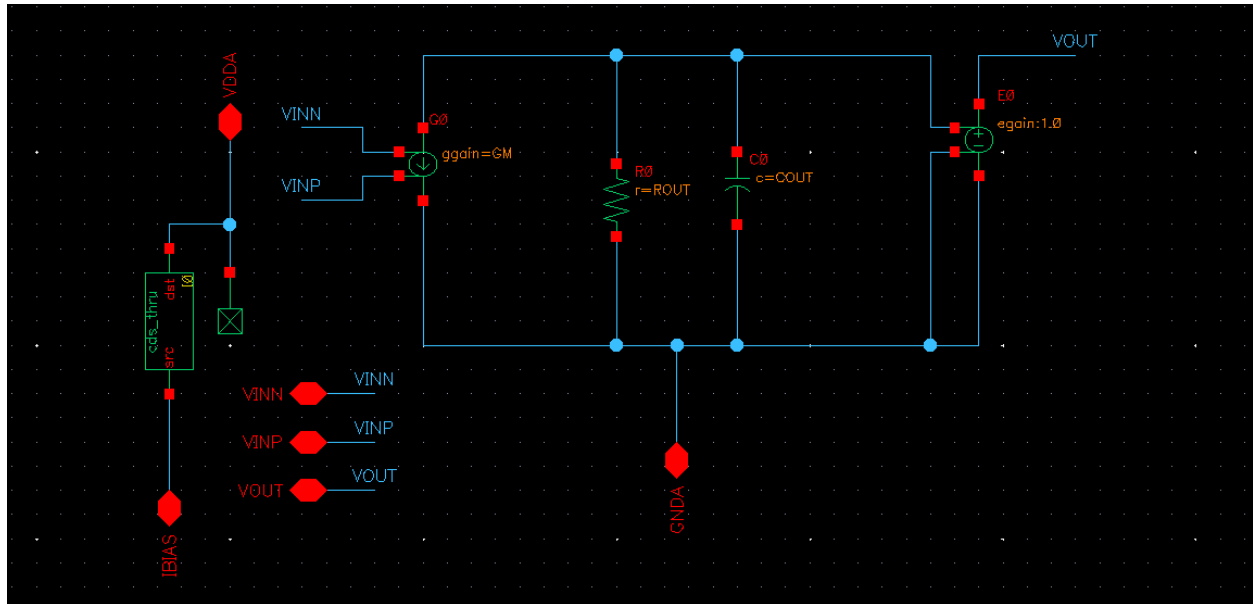


Figure 1:schematic_behav

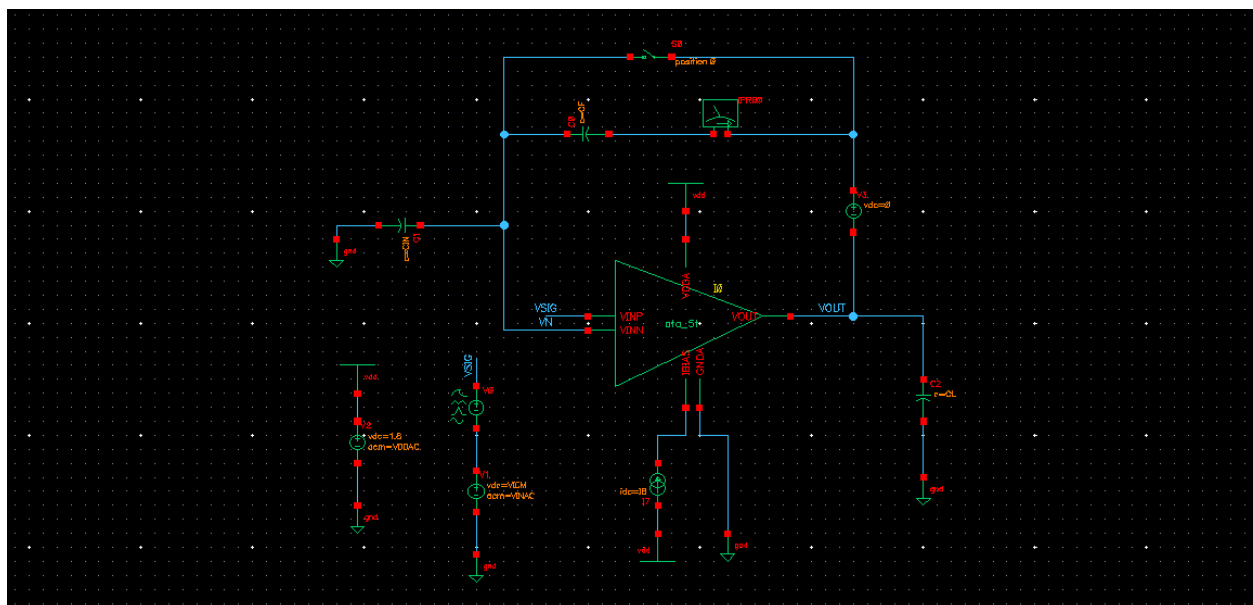


Figure 2: simple non-inverting amplifier



1) Closed loop gain vs frequency.

- Plot V_{out} in dB for the two values of C_{IN} (4pF and 12pF)

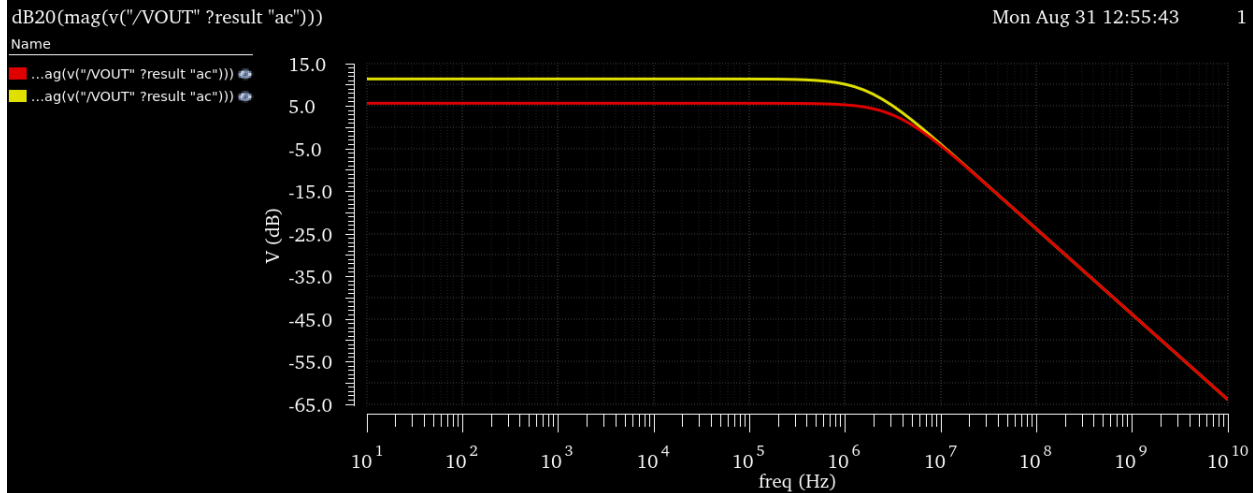


Figure 3: Plot V_{out} in dB for the two values of C_{IN} (4pF and 12pF)

- the DC gain, the bandwidth, and the unity gain frequency

Point	Test	Output	Nominal
Parameters: $C_{IN}=4p$			
1	part2_mosfet:feedback_tb:1	dB20(mag(v(\"/VOUT\" ?result \"ac\")))	
1	part2_mosfet:feedback_tb:1	ymin(dB20(mag(v(\"/VOUT\" ?result \"ac\")))	5.697
1	part2_mosfet:feedback_tb:1	ymin(mag(v(\"/VOUT\" ?result \"ac\")))	1.927
1	part2_mosfet:feedback_tb:1	bandwidth(VF(\"/VOUT\") 3 \"low\")	3.393M
1	part2_mosfet:feedback_tb:1	gainBwProd(VF(\"/VOUT\"))	6.553M
Parameters: $C_{IN}=12p$			
2	part2_mosfet:feedback_tb:1	dB20(mag(v(\"/VOUT\" ?result \"ac\")))	
2	part2_mosfet:feedback_tb:1	ymin(dB20(mag(v(\"/VOUT\" ?result \"ac\")))	11.41
2	part2_mosfet:feedback_tb:1	ymin(mag(v(\"/VOUT\" ?result \"ac\")))	3.718
2	part2_mosfet:feedback_tb:1	bandwidth(VF(\"/VOUT\") 3 \"low\")	1.759M
2	part2_mosfet:feedback_tb:1	gainBwProd(VF(\"/VOUT\"))	6.555M

Figure 4: A_o , BW , W_u



- **Compare the DC gain, BW, and GBW with hand analysis in a table.**

Parameter	Hand Analysis ($C_{IN}=4\text{pF}$)	Simulation ($C_{IN}=4\text{pF}$)	Hand Analysis ($C_{IN}=12\text{pF}$)	Simulation ($C_{IN}=12\text{pF}$)
DC Gain	6.02 dB	5.697 dB	12.04 dB	11.41 dB
Bandwidth (BW)	3.25 MHz	3.393 MHz	1.625 MHz	1.759 MHz
GBW	~ 6.5 MHz	6.553 MHz	~ 6.5 MHz	6.555 MHz

- **Comment**

When C_{IN} is increased from 4pF to 12pF, the closed-loop DC gain doubles (increasing by approximately 6 dB). Because the Gain-Bandwidth Product (GBW) is a constant value determined by the open-loop characteristics of the OTA, the closed-loop bandwidth is proportionally halved (dropping from 3.393 MHz to 1.759 MHz). This perfectly demonstrates the fundamental trade-off in negative feedback systems: any increase in closed-loop gain comes at the direct expense of the bandwidth, while their product (GBW) remains constant.



2) Loop gain vs frequency.

- Plot loop gain in dB for the two values of CIN.

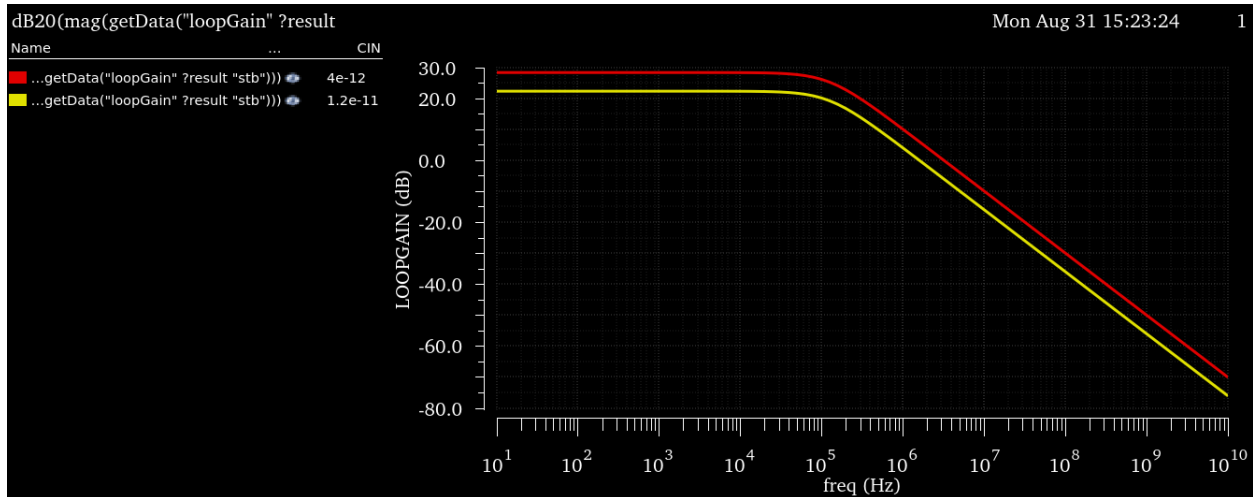


Figure 5: loop gain

- the DC loop gain, the dominant pole, and the unity gain frequency

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	part2_mosfet:feedback_tb:1	dB20(mag(getData("loopGain" ?...	
1	part2_mosfet:feedback_tb:1	ymin(mag(getData("loopGain" ?...	26.35
1	part2_mosfet:feedback_tb:1	unityGainFreq(mag(getData("lo...	3.286M
1	part2_mosfet:feedback_tb:1	ymin(dB20(mag(getData("loop...	28.42
Parameters: CIN=12p			
2	part2_mosfet:feedback_tb:1	dB20(mag(getData("loopGain" ?...	
2	part2_mosfet:feedback_tb:1	ymin(mag(getData("loopGain" ?...	13.18
2	part2_mosfet:feedback_tb:1	unityGainFreq(mag(getData("lo...	1.638M
2	part2_mosfet:feedback_tb:1	ymin(dB20(mag(getData("loop...	22.4

$$\rightarrow \omega_d @ 4p, 12p = \frac{3.286M}{26.35} = 124.7k\text{ohm}$$



- **Compare DC LG and GBW with hand analysis in a table.**

Hand Analysis: Loop Gain and Feedback Factor

To calculate the theoretical Loop Gain (LG), we use the fundamental closed-loop feedback equation:

$$A_{CL} = \frac{A_{OL}}{1 + LG} \Rightarrow LG = A_{OL} \cdot \beta$$

The feedback factor (β) is derived by applying the voltage divider rule at the positive input node (v_p):

$$v_p = v_{out} \left(\frac{Z_{in}}{Z_f + Z_{in}} \right)$$

By substituting the capacitive impedances ($Z_{in} = \frac{1}{sC_{IN}}$ and $Z_f = \frac{1}{sC_F}$), we get:

$$\beta = \frac{v_p}{v_{out}} = \frac{\frac{1}{sC_{IN}}}{\frac{1}{sC_{IN}} + \frac{1}{sC_F}}$$

Simplifying the expression yields:

$$\beta = \frac{C_F}{C_F + C_{IN}}$$

Using $C_F = 4pF$ and the simulated open-loop gain $A_{OL} \approx 52.7 V/V$:

- **Case 1:** $C_{IN} = 4pF$

$$\beta = \frac{4pF}{4pF + 4pF} = 0.5$$

$$LG = A_{OL} \cdot \beta = 52.7 \cdot 0.5 = 26.35 V/V$$



- **Case 2:** $C_{IN} = 12pF$

$$\beta = \frac{4pF}{12pF + 4pF} = 0.25$$

$$LG = A_{OL} \cdot \beta = 52.7 \cdot 0.25 = 13.175 V/V$$

Table: Loop Gain and UGF Comparison

Parameter	Hand Analysis (CIN=4pF)	Simulation (CIN=4pF)	Hand Analysis (CIN=12pF)	Simulation (CIN=12pF)
DC Loop Gain (Linear)	26.35 V/V	26.35 V/V	13.17 V/V	13.18 V/V
DC Loop Gain (dB)	28.41 dB	28.42 dB	22.39 dB	22.40 dB
Unity Gain Frequency	3.275 MHz	3.286 MHz	1.637 MHz	1.638 MHz

Comment: Increasing C_{IN} from 4pF to 12pF decreases the feedback factor (β) from 0.5 to 0.25, causing the feedback signal to experience greater attenuation. Consequently, both the DC loop gain magnitude and the unity gain frequency (UGF) are halved (resulting in a 6 dB drop in DC loop gain), as they are directly proportional to the feedback factor ($LG = A_{OL} \cdot \beta$ and $UGF = GBW \cdot \beta$).



3) Gain Desensitization. Study the variation of closed loop gain with the variation of open loop gain

- Plot closed loop DC gain (magnitude at 10Hz, not dB) vs A_v

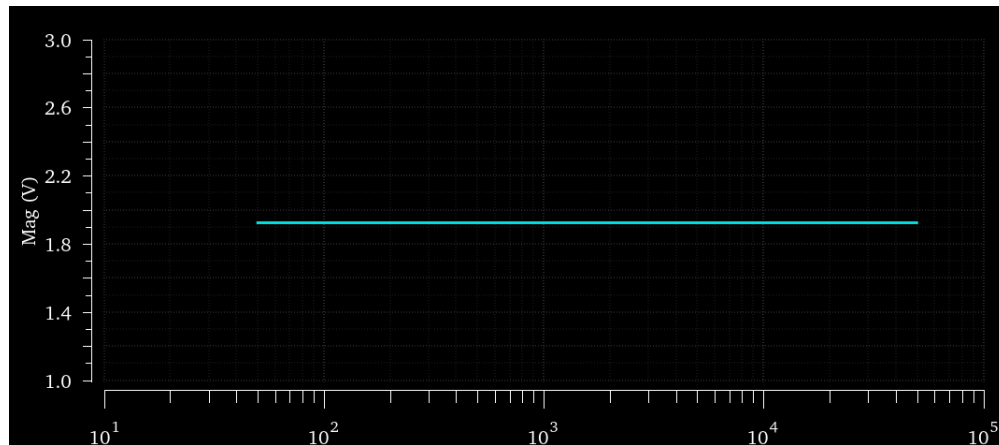


Figure 6: closed loop gain vs. A_v

- Calculate the percent change in closed loop gain

Calculation:

- $A_{CL,initial}$ (at $A_v = 50$) $\approx 1.927 \text{ V/V}$
- $A_{CL,final}$ (at $A_v = 50000$) $\approx 1.927 \text{ V/V}$

$$\text{Percent Change} = \frac{A_{CL,final} - A_{CL,initial}}{A_{CL,initial}} \times 100\%$$

$$\text{Percent Change} = \frac{1.927 - 1.927}{1.927} \times 100\% = 0\%$$

Comment:

Despite the open-loop gain (A_v) experiencing a massive variation of three orders of magnitude (from 50 to 50,000, representing a 60 dB increase), the closed-loop gain remained entirely constant with a 0% change. This clearly demonstrates the gain desensitization property of



negative feedback, which renders the system's overall performance highly stable and immune to drastic variations in the amplifier's internal characteristics.

PART 2: Feedback with Real 5T OTA

1. Closed loop gain vs frequency.

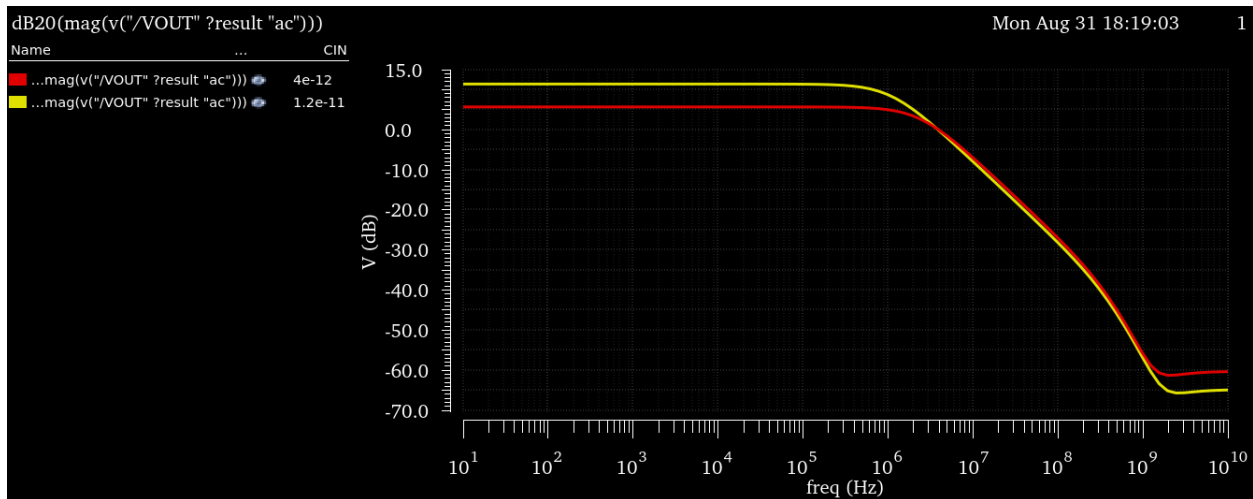


Figure 8: closed loop gain vs. frequency

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	part2_mosfet:feedback_tb:1	dB20(mag(v("/VOUT" ?result "ac...))	
1	part2_mosfet:feedback_tb:1	ymax(dB20(mag(v("/VOUT" ?res...	5.697
1	part2_mosfet:feedback_tb:1	ymax(mag(v("/VOUT" ?result "ac...	1.927
1	part2_mosfet:feedback_tb:1	bandwidth(VF("/VOUT") 3 "low")	2.429M
1	part2_mosfet:feedback_tb:1	gainBwProd(VF("/VOUT"))	4.69M
Parameters: CIN=12p			
2	part2_mosfet:feedback_tb:1	dB20(mag(v("/VOUT" ?result "ac...))	
2	part2_mosfet:feedback_tb:1	ymax(dB20(mag(v("/VOUT" ?res...	11.41
2	part2_mosfet:feedback_tb:1	ymax(mag(v("/VOUT" ?result "ac...	3.718
2	part2_mosfet:feedback_tb:1	bandwidth(VF("/VOUT") 3 "low")	1.103M
2	part2_mosfet:feedback_tb:1	gainBwProd(VF("/VOUT"))	4.11M

Figure 7: Acl, BW, GBW vs. freq



- Compare between the results you obtained here and the results in Part 1 in a table.

Parameter	Part 1 (Behavioral) - 4pF	Part 2 (Real OTA) - 4pF	Part 1 (Behavioral) - 12pF	Part 2 (Real OTA) - 12pF
DC Gain	5.697 dB	5.697 dB	11.41 dB	11.41 dB
Bandwidth	3.393 MHz	2.429 MHz	1.759 MHz	1.103 MHz
GBW	6.553 MHz	4.69 MHz	6.555 MHz	4.11 MHz

Comment:

The bandwidth and GBW obtained using the real 5T OTA are noticeably smaller than those obtained with the ideal behavioral model. The behavioral OTA from Part 1 is a simplified single-pole system that only considers the explicitly defined load and output capacitances. In contrast, the real 5T OTA contains intrinsic transistor parasitic capacitances (such as C_{gs} , C_{gd} , and C_{db}) at all internal nodes and at the output node. These unmodeled parasitic capacitances act as additional loading on the circuit, which lowers the frequency of the dominant and non-dominant poles, ultimately reducing the closed-loop bandwidth and the overall Gain-Bandwidth Product.



2. Loop gain vs frequency.

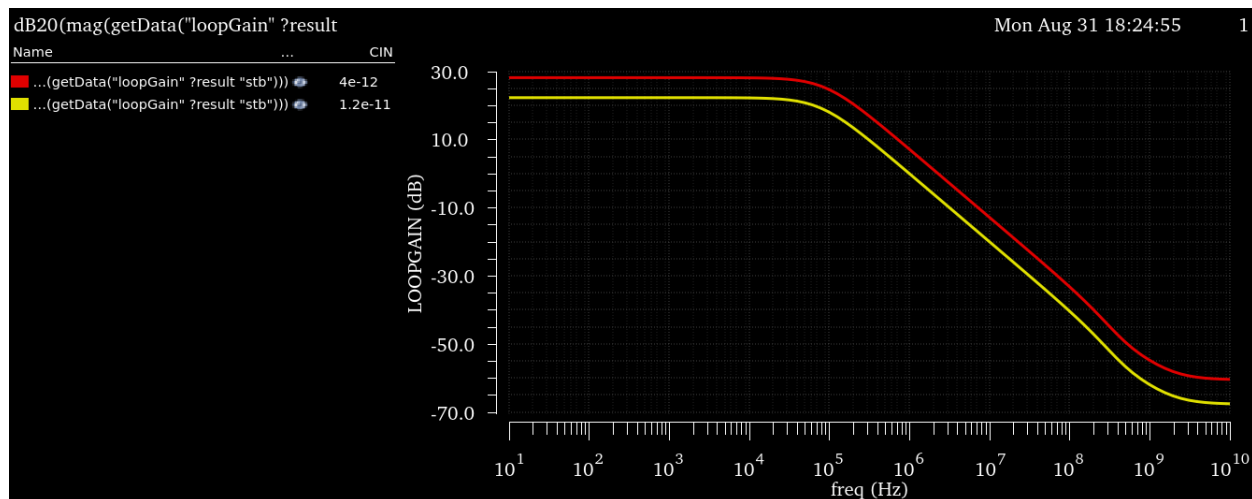


Figure 9: loop gain vs. freq.

Point	Test	Output	Nominal
Parameters: CIN=4p			
1	part2_mosfet:feedback_tb:1	dB20(mag(getData("loopGain" ?...	
1	part2_mosfet:feedback_tb:1	ymin(mag(getData("loopGain" ?...	25.81
1	part2_mosfet:feedback_tb:1	unityGainFreq(mag(getData("lo...	2.354M
1	part2_mosfet:feedback_tb:1	ymin(dB20(mag(getData("loop...	28.24
Parameters: CIN=12p			
2	part2_mosfet:feedback_tb:1	dB20(mag(getData("loopGain" ?...	
2	part2_mosfet:feedback_tb:1	ymin(mag(getData("loopGain" ?...	13.12
2	part2_mosfet:feedback_tb:1	unityGainFreq(mag(getData("lo...	1.023M
2	part2_mosfet:feedback_tb:1	ymin(dB20(mag(getData("loop...	22.36

Figure 10: LG, BW, GBW



- Compare between the results you obtained here and the results in Part 1 in a table.

Parameter	Part 1 (Behavioral) - 4pF	Part 2 (Real OTA) - 4pF	Part 1 (Behavioral) - 12pF	Part 2 (Real OTA) - 12pF
DC Loop Gain	28.42 dB	28.24 dB	22.40 dB	22.36 dB
UGF	3.286 MHz	2.354 MHz	1.638 MHz	1.023 MHz

Comment:

You will notice that the unity gain frequency (UGF) is much smaller than Part 1. This is because the real 5T OTA model in Part 2 includes intrinsic transistor parasitic capacitances (e.g., gate-to-source and drain-to-body capacitances). These unmodeled parasitic elements act as additional loading on the internal nodes, which lowers the frequency of the dominant and non-dominant poles, thereby reducing the available frequency bandwidth of the loop gain.



3. Gain Desensitization.

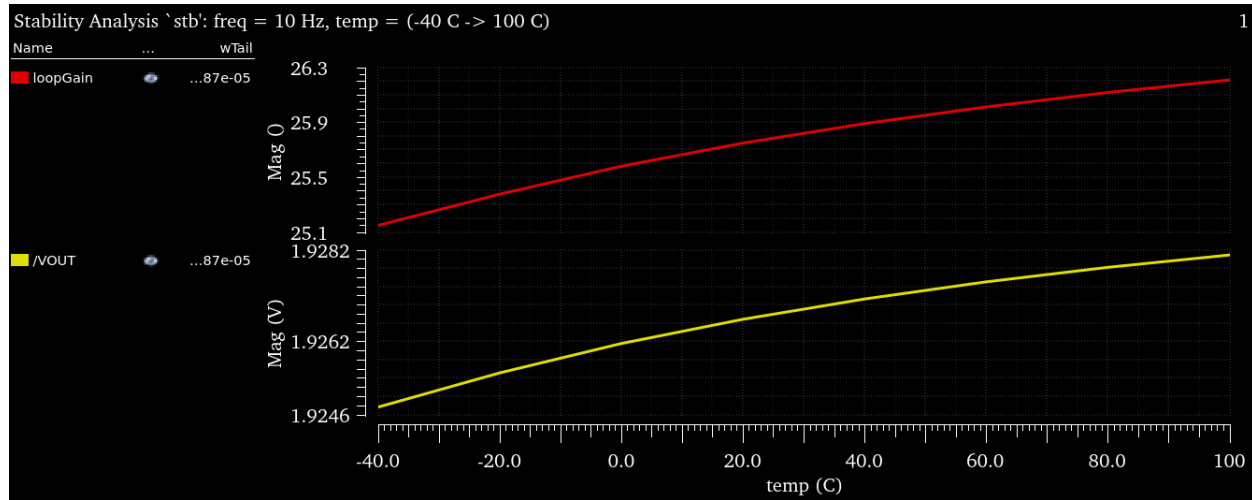


Figure 11: loop gain and closed loop vs. temp

Test	Output	Nominal
part2_mosfet:feedback_tb:1	dB20(mag(v("/VOUT" ?result "ac...	
part2_mosfet:feedback_tb:1	ymax(dB20(mag(v("/VOUT" ?res...	5.703
part2_mosfet:feedback_tb:1	ymax(mag(v("/VOUT" ?result "ac...	1.928
part2_mosfet:feedback_tb:1	dB20(mag(getData("loopGain" ?...	
part2_mosfet:feedback_tb:1	ymax(mag(getData("loopGain" ?...	26.21
part2_mosfet:feedback_tb:1	ymax(dB20(mag(getData("loop...	28.37

Calculations:

- From STB Analysis (DC Loop Gain):

- $LG_{initial}$ (at -40°C) $\approx 25.14 \text{ V/V}$
- LG_{final} (at 100°C) $\approx 26.21 \text{ V/V}$
- $\text{Percent Change} = \frac{26.21 - 25.14}{25.14} \times 100 \approx 4.25\%$



- **From AC Analysis (DC Closed Loop Gain):**

- $A_{CL,initial}$ (at -40°C) $\approx 1.9248 \text{ V/V}$
- $A_{CL,final}$ (at 100°C) $\approx 1.9282 \text{ V/V}$
- $\text{Percent Change} = \frac{1.9282 - 1.9248}{1.9248} \times 100 \approx 0.176\%$

Comment:

Sweeping the temperature from -40°C to 100°C caused the internal loop gain of the OTA to vary by approximately 4.25% due to temperature-dependent variations in transistor mobility and output resistance. However, thanks to the negative feedback mechanism, the closed-loop gain exhibited a negligible variation of only 0.176%. This confirms the gain desensitization property of the feedback loop, rendering the circuit's overall performance highly robust against severe environmental temperature fluctuations.

4. Transient analysis.

- **Case 1: Input Frequency $F_{IN} = 1 \text{ kHz}$**

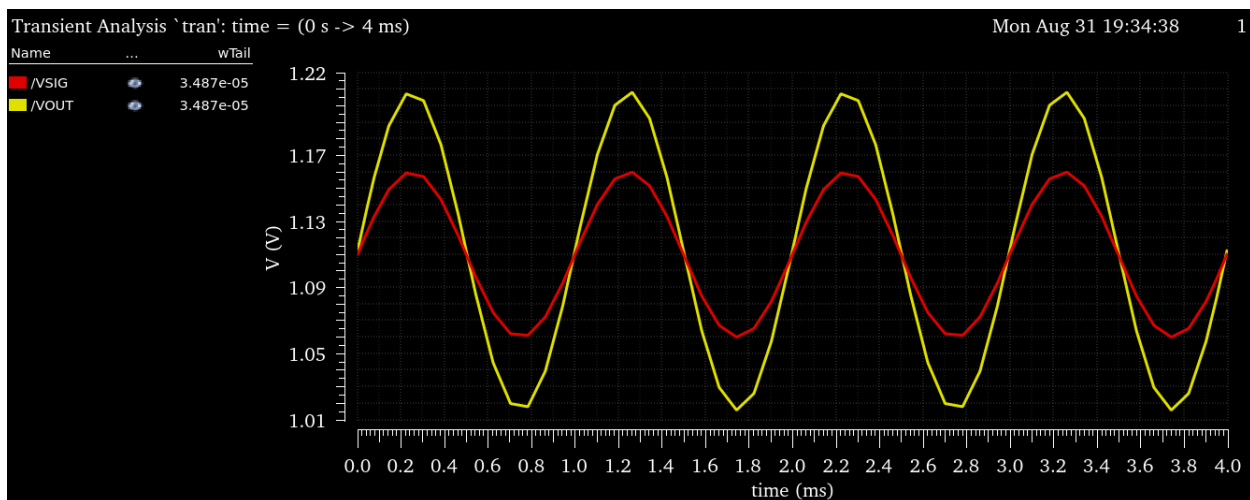


Figure 12: V_{in} and V_{out} vs. time

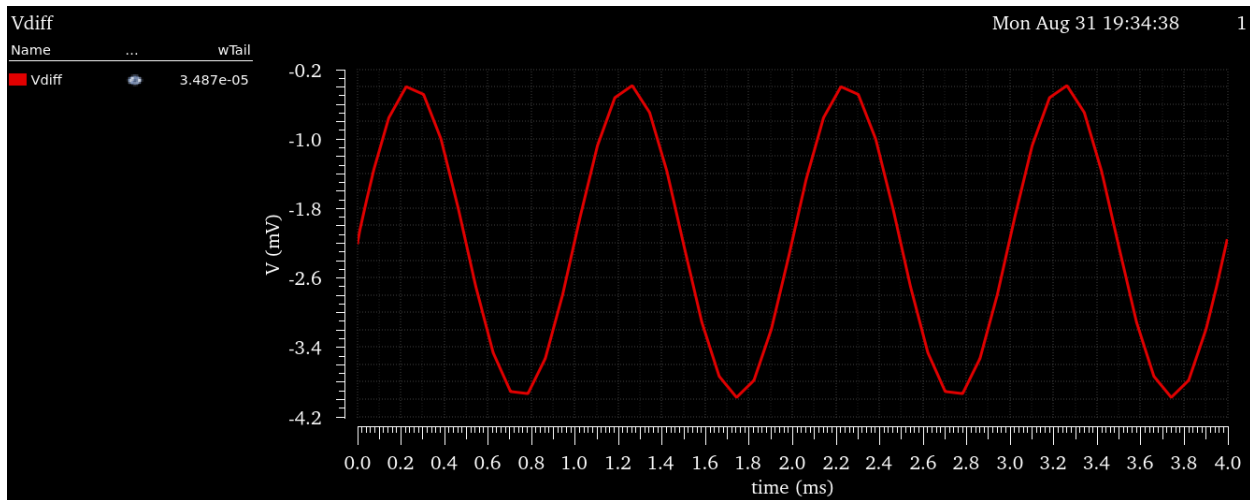


Figure 13: V_{diff} vs. time

- **Peak-to-Peak Calculations:**

- $V_{p-p}(V_{SIG}) \approx 100 \text{ mV}$ (Input signal amplitude is 50mV)
- $V_{p-p}(V_{OUT}) \approx 190 \text{ mV}$ (From the plot, oscillating between $\sim 1.015\text{V}$ and $\sim 1.205\text{V}$)
- $V_{p-p}(V_P - V_N) = V_{p-p}(V_{diff}) \approx 3.6 \text{ mV}$ (From the plot, oscillating between $\sim -4.0\text{mV}$ and -0.4mV)

- **Relation between Output and $(V_P - V_N)$:**

The ratio of the output voltage to the differential input voltage represents the **Open-Loop Gain** (A_{OL}) of the OTA. $A_{OL} = \frac{V_{p-p}(V_{OUT})}{V_{p-p}(V_{diff})} = \frac{190 \text{ mV}}{3.6 \text{ mV}} \approx 52.7 \text{ V/V}$ (Which matches our AC analysis results).

- **Comment:**

At this low frequency (1 kHz), the open-loop gain of the OTA is very high. Therefore, the negative feedback works perfectly to minimize the error signal, creating a "virtual short" at the input (V_{diff} is only 3.6 mV).



The circuit successfully amplifies the input signal by the expected closed-loop DC gain ($190\text{mV}/100\text{mV} \approx 1.9 \text{ V/V}$).

- **Case 2: Input Frequency F_{IN} = Closed-Loop Bandwidth**

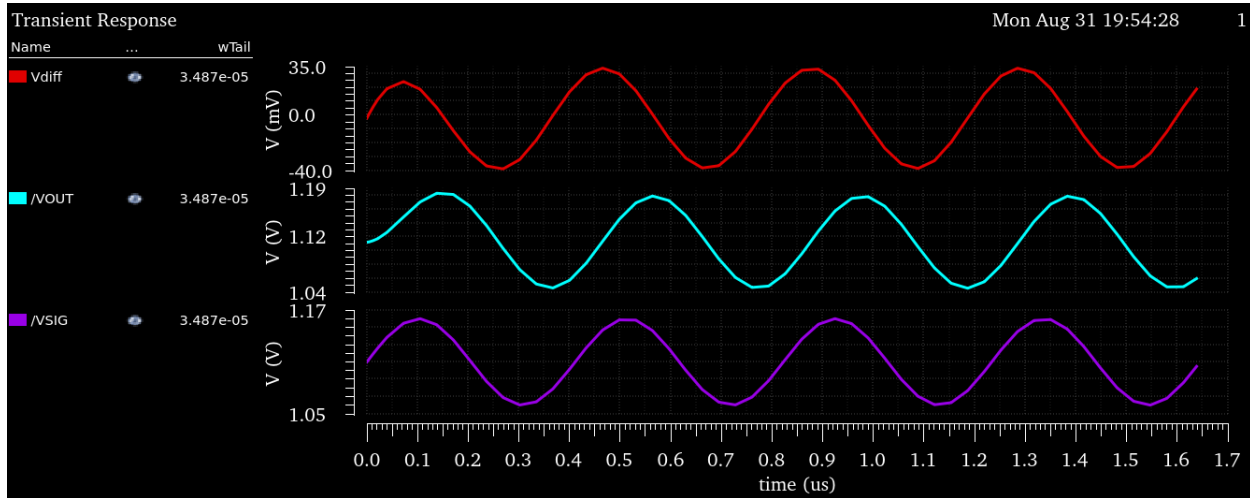


Figure 14: V_{in} , V_{out} and V_{diff} vs. time

- **Peak-to-Peak Calculations:**

- $V_{p-p}(V_{SIG}) \approx 100 \text{ mV}$
- $V_{p-p}(V_{OUT}) \approx 135 \text{ mV}$ (Oscillating between $\sim 1.045\text{V}$ and $\sim 1.18\text{V}$)
- $V_{p-p}(V_P - V_N) = V_{p-p}(V_{diff}) \approx 75 \text{ mV}$ (Oscillating between $\sim -40\text{mV}$ and $+35\text{mV}$)

- **Relation between Output and Input Signal:**

This ratio represents the **Closed-Loop Gain (A_{CL})** at the -3dB cutoff frequency.

$$A_{CL} = \frac{V_{p-p}(V_{OUT})}{V_{p-p}(V_{SIG})} = \frac{135 \text{ mV}}{100 \text{ mV}} = 1.35 \text{ V/V}.$$



(Note: This is approximately $1.92/\sqrt{2}$, which perfectly aligns with the definition of the -3dB bandwidth).

- **Relation between Output and $(V_P - V_N)$:**

This ratio represents the **Open-Loop Gain** (A_{OL}) at this high frequency.

$$A_{OL} = \frac{V_{p-p}(V_{OUT})}{V_{p-p}(V_{diff})} = \frac{135 \text{ mV}}{75 \text{ mV}} = 1.8 \text{ V/V}.$$

Comment:

When operating at a high frequency equal to the closed-loop bandwidth, the OTA's internal open-loop gain drops significantly (from 52.7 down to 1.8). Consequently, the negative feedback loop becomes much less effective. The circuit can no longer maintain the virtual short, causing the differential error signal (V_{diff}) to increase drastically from 3.6 mV to 75 mV. Furthermore, as hinted, the open-loop and closed-loop gains are approximately equal at this frequency ($A_{OL} \approx A_{CL} \approx 1.8$), meaning their asymptotes coincide, and a noticeable phase shift is introduced between the input and output signals.